



Lecture 15

- Laplace Transform

The French Newton Pierre-Simon Laplace (Late 1700)

- Developed mathematics in astronomy, physics, and statistics
- Began work in calculus which led to the Laplace Transform
- Focused later on celestial mechanics
 - One of the first scientists to suggest the existence of black holes



What are Laplace Transforms?

$$F(s) = \int_{0-}^{\infty} f(t) e^{-st} dt$$

$$F(s) = \mathcal{L}[f(t)]$$



$$f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds$$

$$f(t) = \mathcal{L}^{-1}[F(s)]$$

- t is real, s is complex! $s = \sigma + j\omega$
- Assumes $f(t) = 0$ for all $t < 0$
- Note in $f(t) \rightarrow F(s)$, t is integrated and s is variable.
- Conversely, $F(s) \rightarrow f(t)$, t is variable and s is integrated.

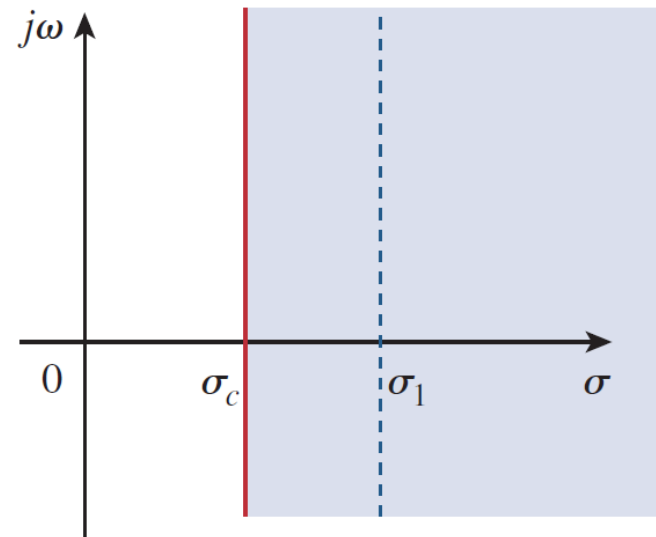
Restrictions

- There are two governing factors that determine whether Laplace transforms can be used:
 - $f(t)$ must be at least piecewise continuous for $t \geq 0$
 - In order for $f(t)$ to have a Laplace transform, the integral must converge to a finite value, The integral converges when

$$\int_{0^-}^{\infty} e^{-\sigma t} |f(t)| dt < \infty$$

Restrictions

- $F(s)$ is undefined outside the region of convergence
- The region of convergence for the Laplace transform is $Re(s) = \sigma > \sigma_c$





Evaluating $F(s) = \mathcal{L}\{f(t)\}$

- Straight Way – do the integral

$$f(t) = 1$$
$$F(s) = \int_0^{\infty} e^{-st} dt = -\frac{1}{s}(0 - 1) = \frac{1}{s}$$

$$f(t) = e^{-at}$$
$$F(s) = \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = \frac{1}{s+a}$$

$$f(t) = \sin t$$
$$F(s) = \int_0^{\infty} e^{-st} \sin(t) dt$$

Integrate by parts



$$f(t) = \sin t$$

$$F(s) = \int_0^{\infty} e^{-st} \sin(t) dt$$

Integrate by parts

$$\int u dv = uv - \int v du$$

$$\int_0^{\infty} e^{-st} \sin(t) dt = -e^{-st} \cos(t) \Big|_0^{\infty} - s \int_0^{\infty} e^{-st} \cos(t) dt =$$

$$1 - s \int_0^{\infty} e^{-st} \cos(t) dt$$

$$\begin{aligned} \int_0^{\infty} e^{-st} \cos(t) dt &= \\ &= -e^{-st} 0 + s \int_0^{\infty} e^{-st} \sin(t) dt \\ &= \frac{1}{1 + s^2} \end{aligned}$$



Evaluating $F(s) = \mathcal{L}\{f(t)\}$

This is the easy way ...

- Recognize a few different transforms
- Learn a few different properties
- Do a little math



Homogeneity and Additivity

$$\mathcal{L}[a_1 f_1(t)] = a_1 \mathcal{L}[f_1(t)] = a_1 F_1(s)$$

$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 \mathcal{L}[f_1(t)] + a_2 \mathcal{L}[f_2(t)] = a_1 F_1(s) + a_2 F_2(s)$$

here a_1 and a_2 are constants

Important implication:

$$\sum_{k=1}^k i_k(t) = 0 \iff \sum_{k=1}^k I_k(s) = 0$$

$$\sum_{k=1}^k u_k(t) = 0 \iff \sum_{k=1}^k U_k(s) = 0$$



Properties: Scaling in Time

$$L\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$$

$$\begin{aligned} L\{f(at)\} &= \int_0^{\infty} f(at) e^{-st} dt = \\ u = at, t &= \frac{u}{a}, dt = \frac{1}{a} du \\ \frac{1}{a} \int_0^{\infty} f(u) e^{-\left(\frac{s}{a}\right)u} du &= \\ \frac{1}{a} F\left(\frac{s}{a}\right) \end{aligned}$$

Example :

$$\begin{aligned} L\{\sin(\omega t)\} &= \frac{1}{\omega} \left(\frac{1}{(s/\omega)^2 + 1} \right) = \\ \frac{1}{\omega} \left(\frac{\omega^2}{s^2 + \omega^2} \right) &= \\ \frac{\omega}{s^2 + \omega^2} \end{aligned}$$



Differentiation

$$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s) - f(0_-)$$

设单边拉普拉斯变换定义为：

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

我们要推导：

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0^+)$$

从定义出发：

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} f'(t)e^{-st} dt$$

这里用分部积分：

$$\int u dv = uv - \int v du$$

令：

$$u = e^{-st}, \quad dv = f'(t) dt$$

于是：

$$du = -se^{-st} dt, \quad v = f(t)$$

代入：

$$\int_0^{\infty} f'(t)e^{-st} dt = [f(t)e^{-st}]_0^{\infty} - \int_0^{\infty} f(t)(-se^{-st}) dt$$

所以：

$$\int_0^{\infty} f'(t)e^{-st} dt = [f(t)e^{-st}]_0^{\infty} + s \int_0^{\infty} f(t)e^{-st} dt$$

后面的积分正好是 $F(s)$ ：

$$s \int_0^{\infty} f(t)e^{-st} dt = sF(s)$$

现在看边界项：

$$[f(t)e^{-st}]_0^{\infty} = \lim_{t \rightarrow \infty} f(t)e^{-st} - f(0^+)$$

在拉普拉斯变换存在的条件下，当 s 足够大时：

$$\lim_{t \rightarrow \infty} f(t)e^{-st} = 0$$

因此：

$$[f(t)e^{-st}]_0^{\infty} = 0 - f(0^+) = -f(0^+)$$

所以：

$$\mathcal{L}\{f'(t)\} = -f(0^+) + sF(s)$$

即：

$$\boxed{\mathcal{L}\{f'(t)\} = sF(s) - f(0^+)}$$



$$\begin{aligned}\mathcal{L}[f^{(n)}(t)] &= s^n F(s) - s^{n-1} f(0_-) - s^{n-2} f^{(1)}(0_-) - \cdots - f^{(n-1)}(0_-) \\ &= s^n F(s) - \sum_{k=0}^{n-1} s^k f^{(n-1-k)}(0_-)\end{aligned}$$

二阶微分也可以从一阶公式推出。

令：

$$g(t) = f'(t)$$

那么：

$$\mathcal{L}\{f''(t)\} = \mathcal{L}\{g'(t)\} = sG(s) - g(0^+)$$

其中：

$$G(s) = \mathcal{L}\{f'(t)\} = sF(s) - f(0^+)$$

所以：

$$\mathcal{L}\{f''(t)\} = s[sF(s) - f(0^+)] - f'(0^+)$$

展开：

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0^+) - f'(0^+)$$

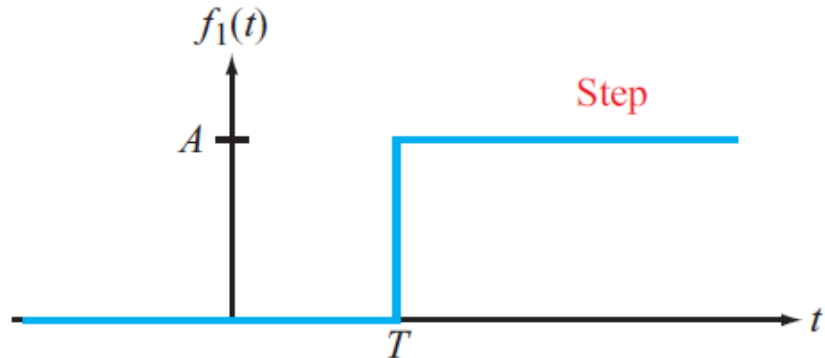


Integration

$$\mathcal{L}\left[\int_{0_-}^t f(\tau)d\tau\right] = \frac{1}{s}F(s)$$

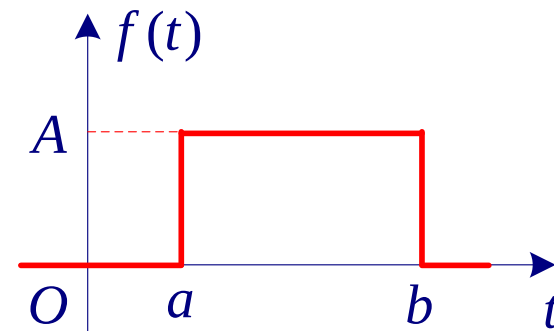
Time Shift

$$\mathcal{L}[f(t-\tau)] = e^{-s\tau} F(s)$$



$$\begin{aligned} F_1(s) &= \int_{0^-}^{\infty} f_1(t) e^{-st} dt = \int_{0^-}^{\infty} A u(t-T) e^{-st} dt \\ &= A \int_T^{\infty} e^{-st} dt = -\frac{A}{s} e^{-st} \Big|_T^{\infty} = \frac{A}{s} e^{-sT}. \end{aligned}$$

Example



$$f(t) = A[u(t-a) - u(t-b)]$$

$$F(s) = A\mathcal{L}[u(t-a) - u(t-b)] = \frac{A}{s} (e^{-as} - e^{-bs})$$



Frequency Shift

$$\mathcal{L}[e^{\alpha t} f(t)] = F(s - \alpha)$$

- Example

$$\mathcal{L}[e^{-\alpha t} \sin \omega t] = \frac{\omega}{(s + \alpha)^2 + \omega^2}$$



TABLE 12.2 An Abbreviated List of Operational Transforms

Operation	$f(t)$	$F(s)$
Multiplication by a constant	$Kf(t)$	$KF(s)$
Addition/subtraction	$f_1(t) + f_2(t) - f_3(t) + \dots$	$F_1(s) + F_2(s) - F_3(s) + \dots$
First derivative (time)	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Second derivative (time)	$\frac{d^2f(t)}{dt^2}$	$s^2F(s) - sf(0^-) - \frac{df(0^-)}{dt}$
n th derivative (time)	$\frac{d^nf(t)}{dt^n}$	$s^nF(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt}$ $- s^{n-3}\frac{d^2f(0^-)}{dt^2} - \dots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}$
Time integral	$\int_0^t f(x) dx$	$\frac{F(s)}{s}$
Translation in time	$f(t - a)u(t - a), a > 0$	$e^{-as}F(s)$
Translation in frequency	$e^{-at}f(t)$	$F(s + a)$
Scale changing	$f(at), a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
First derivative (s)	$tf(t)$	$-\frac{dF(s)}{ds}$
n th derivative (s)	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s integral	$\frac{f(t)}{t}$	$\int_s^\infty F(u) du$



TABLE 12.1 An Abbreviated List of Laplace Transform Pairs

Type	$f(t)$ ($t > 0-$)	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$



Find $F(s)$ given that

$$f(t) = \begin{cases} 5t, & 0 < t < 1 \\ -5t, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases}$$



15.9 Determine the Laplace transforms of these functions:

(a) $f(t) = (t - 4)u(t - 2)$

(b) $g(t) = 2e^{-4t}u(t - 1)$

(c) $h(t) = 5 \cos(2t - 1)u(t)$

(d) $p(t) = 6[u(t - 2) - u(t - 4)]$



The Inverse Laplace Transform

- $F(s) = \frac{N(s)}{D(s)}$
- Steps to Find the Inverse Laplace Transform:
 - 1. Decompose $F(s)$ into simple terms using partial fraction expansion.
 - 2. Find the inverse of each term by matching entries in the table.



Simple Poles

$$F(s) = \frac{N(s)}{(s + p_1)(s + p_2) \cdots (s + p_n)} \quad p_i \neq p_j \text{ for all } i \neq j$$

$$F(s) = \frac{k_1}{s + p_1} + \frac{k_2}{s + p_2} + \cdots + \frac{k_n}{s + p_n}$$

$$(s + p_1)F(s) = k_1 + \frac{(s + p_1)k_2}{s + p_2} + \cdots + \frac{(s + p_1)k_n}{s + p_n}$$

$$k_i = (s + p_i)F(s) \big|_{s=-p_i}$$

$$f(t) = (k_1 e^{-p_1 t} + k_2 e^{-p_2 t} + \cdots + k_n e^{-p_n t})u(t)$$

Repeated Poles

- Suppose $F(s)$ has n repeated poles

$$F(s) = \frac{k_n}{(s+p)^n} + \frac{k_{n-1}}{(s+p)^{n-1}} + \cdots + \frac{k_2}{(s+p)^2} + \frac{k_1}{s+p} + F_1(s)$$

Determine coefficient k_n $k_n = (s+p)^n F(s) \big|_{s=-p}$

$$k_{n-1} = \frac{d}{ds}[(s+p)^n F(s)] \big|_{s=-p}$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+a)^n}\right] = \frac{t^{n-1}e^{-at}}{(n-1)!}u(t)$$

$$k_{n-m} = \frac{1}{m!} \frac{d^m}{ds^m}[(s+p)^n F(s)] \big|_{s=-p}$$

$$f(t) = \left(k_1 e^{-pt} + k_2 t e^{-pt} + \frac{k_3}{2!} t^2 e^{-pt} + \cdots + \frac{k_n}{(n-1)!} t^{n-1} e^{-pt} \right) u(t) + f_1(t)$$



Complex Poles

$$F(s) = \frac{A_1s + A_2}{s^2 + as + b} + F_1(s)$$

$$s^2 + as + b = (s + \alpha)^2 + \beta^2$$

$$A_1s + A_2 = A_1(s + \alpha) + B_1\beta$$

$$F(s) = \frac{A_1(s + \alpha)}{(s + \alpha)^2 + \beta^2} + \frac{B_1\beta}{(s + \alpha)^2 + \beta^2} + F_1(s)$$

$$f(t) = (A_1e^{-\alpha t} \cos \beta t + B_1e^{-\alpha t} \sin \beta t)u(t) + f_1(t)$$



Exercise

$$F(s) = \frac{s^2 + 3s + 5}{s^3 + 6s^2 + 11s + 6}$$

$$F(s) = \frac{s^2 + 3s + 5}{(s+1)(s+2)(s+3)} = \frac{K_1}{s+1} + \frac{K_2}{s+2} + \frac{K_3}{s+3}$$



Exercise

$$F(s) = \frac{10s^2 + 4}{s(s+1)(s+2)^2}$$

$$F(s) = \frac{K_{11}}{s} + \frac{K_{21}}{s+1} + \frac{K_{31}}{s+2} + \frac{K_{32}}{(s+2)^2}$$

$$f(t) = [1 - 14e^{-t} + (13 + 22t)e^{-2t}] \varepsilon(t)$$



The Convolution Integral

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda) d\lambda$$

or simply

$$y(t) = x(t) * h(t)$$

$$F(s) = \mathcal{L}[f_1(t) * f_2(t)] = F_1(s)F_2(s)$$